Recursive Bipartite Matching Distance for Game Tree Abstraction with Bounded Error

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Abstract

We introduce the *recursive bipartite matching* (RBM) distance, a metric on rooted trees defined by recursively solving minimum-cost bipartite matchings over children. Unlike scalar hand-strength metrics used in prior game abstraction work, RBM captures the full strategic structure of continuation trees. We prove three results: (1) RBM is a metric when the leaf distance is a metric; (2) merging two trees at distance ε introduces expected-value error at most $\varepsilon/2$; and (3) an online learner that clusters game states by RBM distance achieves cumulative regret $O(KV_{\max} + T\varepsilon/2)$ with fixed threshold, or $O(\sqrt{T})$ under ε-annealing. On Rhode Island Hold'em, RBM-based abstraction beats Earth Mover's Distance abstraction 7–0 across compression levels measured by EV error, and counterfactual regret minimization on RBM-compressed games achieves zero exploitability at natural cluster boundaries.

1 Introduction

Abstraction is the dominant paradigm for solving large extensive-form games. The idea is straightforward: collapse strategically similar game states into equivalence classes, solve the smaller *abstract* game, and map the abstract strategy back to the original. Superhuman poker agents [Bowling et al., 2015, Moravčík et al., 2017, Brown and Sandholm, 2018] all rely on some form of abstraction to make computation tractable.

The quality of an abstraction depends critically on the *distance metric* used to decide which states are "similar." The standard approach in poker AI, pioneered by Gilpin and Sandholm [2007], clusters information sets using Earth Mover's Distance (EMD) on hand-strength histograms: distributions over scalar equity values obtained by evaluating each hand against a uniform random opponent. This is effective but inherently limited—it reduces each game state to a one-dimensional summary, discarding the branching structure of future play.

Two hands may have identical equity distributions yet lead to fundamentally different strategic situations. A made hand and a draw can have the same expected value but completely different continuation trees: one calls safely, the other bets aggressively to deny correct pot odds or folds against reraises. Scalar metrics cannot distinguish them.

We propose a structural alternative. The *recursive bipartite matching* (RBM) distance between two game trees is defined by recursively computing minimum-cost bipartite matchings between their children at every level. Leaves are compared by payoff difference; internal nodes are compared by the cost of optimally aligning their subtrees. Structural mismatches (differing numbers of children) incur an explicit phantom penalty. The result is a metric that respects both the values at the leaves and the *shape* of the computation that produces them.

This paper develops the theory and practice of RBM-based game abstraction in three parts.

*Work in progress. Code: `https://github.com/struktured-labs/recursive-bipartite-matching`

The metric and its error bound. We prove that RBM is a metric (Section 2), and that merging two trees at distance ε introduces EV error at most $\varepsilon/2$ (Section 3). The bound is tight at leaves but strictly tighter at branching nodes, where averaging reduces error.

The location problem. Using a structurally rich distance creates a compressed *EV graph*—a DAG of merged game states—that faithfully preserves strategic content. But locating oneself in this graph during live play requires comparing one's continuation tree against cluster representatives, which can be as expensive as solving the uncompressed game (Section 4).

The online learning resolution. We dissolve the location problem by building the EV graph *incrementally through self-play.* The learner always knows where it is because it constructed the graph from its own positions. We prove that this online learner achieves bounded cumulative regret, with sublinear average regret under ε-annealing (Section 5).

Experiments on Rhode Island Hold'em [Gilpin and Sandholm, 2005] (Section 6) validate the theory: RBM-based clustering beats EMD-based clustering 7–0 across compression levels on EV error, CFR on compressed games achieves zero exploitability at natural cluster boundaries, and the online learner converges to stable clusters within 100 games with 99.5% cache hit rates.

2 Recursive Bipartite Matching Distance

2.1 Definition

Let $\mathcal{T}$ denote the space of finite rooted trees with real-valued leaves. We define a distance $d : \mathcal{T} \times \mathcal{T} \to \mathbb{R}_{\geq 0}$ recursively.

Definition 2.1 (RBM Distance)**.** *Given two rooted trees T_1 and T_2:*

Base case. *If both are leaves with values v_1, v_2:*

$$d(\text{leaf}_1, \text{leaf}_2) = |v_1 - v_2|.$$

Recursive case. *Let n_1 have children $\{c_1^1, \ldots, c_1^m\}$ and n_2 have children $\{c_2^1, \ldots, c_2^n\}$, with $m \leq n$ (WLOG).*

1. *Compute the pairwise cost matrix W where $W[i,j] = d(\text{sub}(c_1^i), \text{sub}(c_2^j))$.*
2. *Pad the smaller child set with $|n - m|$* phantom nodes*. The cost of matching a real subtree S against a phantom is $\delta(S)$.*
3. *Solve the minimum-cost perfect matching on the padded bipartite graph via the Hungarian method [Kuhn, 1955], yielding optimal matching M^*.*
4. *$d(T_1, T_2) = \text{cost}(M^*)$.*

The phantom penalty δ quantifies the cost of structural mismatch. The cleanest formulation is $\delta(S) = d(S, \emptyset)$, treating the phantom as a designated empty tree in the metric space. Under this choice, all conditions needed for the metric proof are consequences of the triangle inequality at lower depth (see Remark 2.3).

Children are treated as *unordered sets.* The matching finds the best structural alignment without requiring a canonical ordering—exactly the right semantics for game trees, where children of a chance node (possible deals) or decision node (possible actions) have no inherent order.

2.2 Metric Properties

Theorem 2.2 (RBM is a Metric)**.** *Let d_L be a metric on leaf values and let $\delta(S) = d(S, \emptyset)$ for a designated empty tree $\emptyset$. Then the RBM distance d is a metric on $\mathcal{T}$.*

Proof sketch. By structural induction on tree depth. The base case ($k = 0$) reduces to d_L, which is a metric by assumption.

Identity: The identity matching $\{(c_i, c_i)\}$ has cost zero by induction; since all costs are nonneg, the minimum-cost matching achieves $d(T, T) = 0$.

Symmetry: The cost matrix for $d(T_2, T_1)$ is the transpose of that for $d(T_1, T_2)$, using symmetry at depth $< k$. The minimum-cost perfect matching is invariant under transposition.

Triangle inequality: Given optimal matchings M_{12} (between T_1, T_2) and M_{23} (between T_2, T_3), compose through T_2 to construct a feasible matching M_{13} between T_1 and T_3. Children "through-matched" via some $b_j \in T_2$ satisfy $d(\mathrm{sub}(a_i), \mathrm{sub}(c_k)) \leq d(\mathrm{sub}(a_i), \mathrm{sub}(b_j)) + d(\mathrm{sub}(b_j), \mathrm{sub}(c_k))$ by induction. Dangling children (matched to phantoms on one side) are handled by the 1-Lipschitz property of δ. Summing over all pairs: $\mathrm{cost}(M_{13}) \leq \mathrm{cost}(M_{12}) + \mathrm{cost}(M_{23})$. Since $d(T_1, T_3) \leq \mathrm{cost}(M_{13})$ by optimality, the triangle inequality follows.

The full proof appears in Appendix A. □

Remark 2.3 (Phantom Penalty)**.** *With $\delta(S) = d(S, \emptyset)$, the phantom triangle property $d(S_1, S_2) \leq \delta(S_1) + \delta(S_2)$ and the 1-Lipschitz condition $\delta(S_1) \leq d(S_1, S_2) + \delta(S_2)$ are both instances of the triangle inequality. The theorem is self-contained: if the leaf distance is a metric, then RBM with empty-tree phantoms is a metric.*

2.3 Relation to Prior Tree Distances

RBM is related to but distinct from tree edit distance [Zhang, 1996], which uses insert/delete/relabel operations on ordered trees. Edit distance is more flexible but less natural for *unordered* trees where children are interchangeable. The recursive matching structure also connects to hierarchical optimal transport [Mémoli, 2011] and the Tree Mover's Distance [Chuang et al., 2022], but those works target different applications (shape matching and GNN analysis, respectively). See Section 7 for a detailed comparison.

3 Game Tree Compression via Merge

3.1 The Merge Operation

The RBM distance induces a natural *merge* operation. Given trees T_1, T_2 with small distance, produce a representative T^* that captures their shared structure.

Definition 3.1 (Tree Merge)**.** *Given the optimal matching M^* between children of T_1 and T_2:*

- ***Matched pairs*** *$(c_1^i, c_2^j) \in M^*$: recursively merge their subtrees.*
- ***Phantom-matched children****: either retain at reduced weight or drop (lossy compression).*
- ***Leaf merging****: average the payoff values: $v^* = (v_1 + v_2)/2$.*

3.2 The EV Graph

Repeated merging across all subtrees at each depth produces a directed acyclic graph—the *EV graph*. Multiple original game paths converge to the same node, meaning "from here, these situations play out essentially the same way." This is hierarchical clustering over subtrees using RBM as the metric, with each cluster represented by a merged prototype tree.

3.3 EV Error Bound

Theorem 3.2 (EV Error Bound Under Merging)**.** *Let T_1, T_2 be rooted trees with real-valued leaves, $d(T_1, T_2) = \varepsilon$. Let $T^* = \mathrm{merge}(T_1, T_2)$ with equal-weight leaf averaging. Then:*

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_i)| \leq \frac{\varepsilon}{2}, \quad i \in \{1, 2\}.$$

Proof sketch. By structural induction on tree depth.

Base case (leaves): $T^* = \mathrm{leaf}((v_1 + v_2)/2)$, so $|\mathrm{EV}(T^*) - v_1| = |v_2 - v_1|/2 = \varepsilon/2$.

Inductive case: Let M^* be the optimal matching with matched-pair costs d_i and total $\varepsilon = \sum_i d_i + \sum_{\text{phantom}} \delta(b_j)$. Each merged child satisfies $|\mathrm{EV}(c_i^*) - \mathrm{EV}(\mathrm{sub}(a_i))| \leq d_i/2$ by induction. With uniform weighting over m children:

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_1)| \leq \frac{1}{m} \sum_{i=1}^{m} \frac{d_i}{2} = \frac{1}{2m} \sum_{i=1}^{m} d_i \leq \frac{\varepsilon}{2}.$$

For trees with branching factor $m > 1$ and no phantom terms, the bound tightens to $\varepsilon/(2m) \leq \varepsilon/4$. The stated bound $\varepsilon/2$ is tight only at leaves or degenerate single-child chains. □

Corollary 3.3 (Weighted Merges)**.** *For unequal-weight merges with weights $w_1 + w_2 = 1$:*

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_i)| \leq w_{3-i} \cdot \varepsilon.$$

Equal weights $w_1 = w_2 = 1/2$ recover $\varepsilon/2$.

Corollary 3.4 (Iterated Merges)**.** *When k trees $T_1, \ldots, T_k$ are merged into a single T^* via successive pairwise merges:*

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_i)| \leq \max_j d(T_i, T_j) \leq \varepsilon_{\text{diam}},$$

where $\varepsilon_{\text{diam}}$ is the cluster diameter.

4 The Location Problem

Building the EV graph offline is one challenge; *finding yourself in it during live play* is another.

4.1 Statement

Given a compressed EV graph G and a game state v (known: cards, history), find the corresponding node $\phi(v) \in G$.

In flat abstraction, location is trivial: precompute a lookup table mapping each state to its bucket, $O(1)$ at query time. With tree-based abstraction, the equivalence class of v depends on its *entire continuation subtree*—the same non-local structure that makes the distance expressive.

4.2 Complexity Barrier

Proposition 4.1 (Location Cost)**.** *Let b be the branching factor, h the remaining depth, K the number of clusters. Naive location requires $O(K \cdot b^{2h})$ time per game step, since each distance computation $T(h)$ satisfies $T(h) = b^2 \cdot T(h-1) + O(b^3)$, giving $T(h) = O(b^{2h})$.*

Algorithm 1 Online EV Graph Construction via Self-Play

```
Initialize EV graph G ← ∅, threshold ε > 0
for each game t = 1, 2, ..., T do
  for each decision point in game t do
    Observe game state s; build partial subtree T_s
    Find nearest cluster: c* = arg min_c d(T_s, rep(c))
    if d(T_s, rep(c*)) ≤ ε (cache hit) then
      Use cluster c*'s strategy; record outcome
    else {(cache miss)}
      Create new cluster from T_s; explore
    end if
  end for
  Update cluster representatives with observed payoffs
  Merge clusters where d(rep(c_i), rep(c_j)) < ε_merge
  Optionally anneal ε downward
end for
```

Over a full game of d steps, the total cost is $O(d \cdot K \cdot n^2)$ where $n = b^h$ is the number of leaves. This can exceed the cost of solving the uncompressed game, defeating the purpose of compression. The location problem likely admits a lower bound of $\Omega(n)$ per step, because the equivalence classes depend on global subtree structure defined by optimal (Hungarian) matchings.

This reveals a fundamental tradeoff in game abstraction: *the more structurally faithful the abstraction, the harder it is to navigate at runtime.*

5 Online Learning with Bounded Regret

The offline approach—enumerate, cluster, then look up—hits the location problem wall. We propose a more natural formulation: build the EV graph *dynamically through play.*

5.1 The Online EV Graph Learner

Algorithm 1 resolves the core tensions:

The location problem dissolves. The learner never needs to locate itself in a precomputed graph because it built the graph *from* its own positions. Each cluster knows its game states because it was born from one.

The enumeration problem dissolves. Only states actually encountered are represented. The graph concentrates resolution where it matters—the same insight behind MCTS [Coulom, 2006].

The error bound becomes a regret bound. When the learner uses cluster c^*'s strategy on a state at distance $\leq \varepsilon$, the EV loss is bounded by $\varepsilon/2$ (Theorem 3.2).

5.2 Regret Bound

Theorem 5.1 (Online Regret Bound)**.** *Consider the online EV graph learner (Algorithm 1) with threshold $\varepsilon > 0$, playing T games against a stationary environment. Let K be the number of distinct*

strategic equivalence classes (game trees up to ε-distance), and $V_{\max} = \max_t |\mathrm{EV}(T_t)|$. Then:

1. ***Exploration bound:*** *Total cache misses $\leq K$.*
2. ***Per-step exploitation error:*** *At most $\varepsilon/2$ per cache hit.*
3. ***Cumulative regret:***
$$R(T) \leq K \cdot V_{\max} + (T - K) \cdot \frac{\varepsilon}{2}.$$
4. ***Average regret:*** *$R(T)/T \to \varepsilon/2$ as $T \to \infty$.*
5. ***ε-annealing:*** *With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$, $R(T) = O(\sqrt{T})$ and $R(T)/T \to 0$.*

Proof sketch. Part 1: Each equivalence class triggers at most one cache miss. Once representative T_t is created, all future trees in the same class satisfy $d(T_{t'}, T_t) \leq \varepsilon$, producing cache hits.

Parts 2–3: Cache hits incur error $\leq \varepsilon/2$ (Theorem 3.2). Cache misses incur at most $V_{\max}$. Summing: $R(T) \leq KV_{\max} + (T - K)\varepsilon/2$.

Part 5: With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$, exploitation regret sums to $\sum_{t=1}^{T} \varepsilon_0/(2\sqrt{t}) \leq \varepsilon_0\sqrt{T}$. Exploration cost is at most $N \cdot V_{\max}$ (constant). Total: $R(T) = O(\sqrt{T})$.

The full proof appears in Appendix C. □

5.3 Connection to Lipschitz Bandits

The $O(\sqrt{T})$ rate matches the minimax-optimal rate for multi-armed bandits [Auer et al., 2002]. The structural advantage over generic bandits is that RBM provides a *geometry* on the arm space: nearby clusters have similar payoffs by the EV error bound. This is analogous to Lipschitz bandits or continuum-armed bandits [Kleinberg, 2005], where metric structure enables faster learning. In the EV graph setting, the metric is not assumed but *derived* from the game structure via recursive bipartite matching.

6 Experiments

We evaluate on Rhode Island Hold'em [Gilpin and Sandholm, 2005], a simplified poker variant designed for AI research: 2 players, 1 hole card each, 2 community cards (flop + turn), 3 betting rounds with limit betting, and 3-card hand rankings (trips > straight > flush > pair > high card).

All experiments use a configurable deck (2–4 ranks, 4 suits). The implementation is in OCaml using Jane Street Core libraries, with an $O(n^3)$ Hungarian algorithm for bipartite matching.

6.1 RBM vs EMD Comparison

We compare three distance metrics for game abstraction:

- **RBM**: recursive bipartite matching on full game trees (∼1141 nodes per deal),
- **EMD**: Earth Mover's Distance on showdown outcome distributions (win/draw/lose probabilities), the standard poker AI metric [Gilpin and Sandholm, 2007],
- **Scalar EV**: absolute difference in expected values.

For each metric, we compute pairwise distances on 25 sampled deals (3-rank deck, fixed P1 card, varied opponent/community), then perform agglomerative clustering at multiple ε thresholds. At each compression level (cluster count), we measure the maximum EV error across clusters.

Table 1 shows results. **RBM wins 7–0** across all non-trivial compression levels (ties only at the trivial $k = 1$). The result is striking: RBM achieves *zero* EV error at 8.3× compression ($k = 3$), while EMD (Gilpin & Sandholm's metric) incurs 30.28 error even at $k = 4$ clusters—**over 30× worse**. The root cause: EMD groups deals by community-card showdown distribution (same community = same win/lose/draw histogram regardless of opponent), merging strategically

Table 1: RBM vs EMD vs Scalar EV: maximum EV error at matched cluster counts (3-rank Rhode Island Hold'em, 25 sampled deals with varied community cards). Lower is better. RBM achieves **zero error** down to $k = 3$ clusters while EMD has 30.28 error even at $k = 4$.

Clusters	Compress	RBM Error	EMD Error	EV Error	Winner
25	1.0×	0.00	30.28	30.28	RBM
15	1.7×	0.00	30.28	30.28	RBM
10	2.5×	0.00	30.28	30.28	RBM
7	3.6×	0.00	30.28	30.28	RBM
5	5.0×	0.00	30.28	30.28	RBM
3	8.3×	0.00	30.28	30.28	RBM
2	12.5×	25.96	34.07	34.07	RBM
1	25.0×	54.51	54.51	54.51	tie

Table 2: CFR exploitability on full vs. RBM-compressed Rhode Island Hold'em (3-rank, 2 betting rounds, 1000 CFR iterations). RBM compression to 3 clusters achieves **zero exploitability**—the abstraction is strategically lossless.

ε	Clusters	EV Error	Exploitability
(full)	10	—	0.0000
0	3	0.0000	0.0000
50	3	0.0000	0.0000
100	3	0.0000	0.0000
200	3	0.0000	0.0000
500	3	0.0000	0.0000
1000	1	7.0926	5.6573

different hands. RBM groups by actual game tree structure, correctly separating wins, losses, and draws.

6.2 CFR on Compressed Games

We run counterfactual regret minimization (CFR) [Zinkevich et al., 2007] on both the full game and RBM-compressed abstractions to measure exploitability—the gap between the computed strategy and Nash equilibrium.

Table 2 reveals a striking pattern. RBM compression from 10 information sets to just 3 clusters (a 3.3× reduction) achieves *exactly zero* exploitability across all ε up to 500—the abstraction is **strategically lossless**. The RBM distance has correctly identified the 3 natural equivalence classes (different ranks relative to the board). Only when forced to merge all hands into a single cluster ($\varepsilon = 1000$) does exploitability spike to 5.66, consistent with the EV error bound of Theorem 3.2.

6.3 Multi-Scale Results

Table 3 shows RBM vs EMD across three game scales. RBM dominates at every scale, achieving zero error at aggressive compression levels where EMD retains substantial error.

Table 3: RBM vs EMD across game scales (30 sampled deals per scale, varied community cards). RBM wins column counts victories across 8 target compression levels. RBM achieves zero error down to $k = 2$ at all scales.

Scale	Cards	RBM wins	EMD wins	Ties	RBM zero-err k
3-rank	12	7	0	1	2
4-rank	16	7	0	1	2
5-rank	20	7	0	1	2
Total		**21**	**0**	**3**	

Table 4: Online learner convergence on 2-rank Rhode Island Hold'em. With random community cards, the learner discovers the natural cluster structure by game 100 and achieves near-perfect cache hit rates by game 600.

Games	Clusters	Cache Hit Rate
100	3	95.0%
200	3	97.5%
400	3	98.8%
600	3	99.5%

6.4 Metric Property Verification

We verify RBM metric properties exhaustively on the 4-rank game (91 deal trees, 753,571 triples for the triangle inequality):

- **Identity**: $d(T, T) = 0$ for all 91 trees. ✓
- **Symmetry**: $d(T_i, T_j) = d(T_j, T_i)$ for all 4095 pairs. ✓
- **Triangle inequality**: verified on all 753,571 triples. ✓

6.5 Online Learner Convergence

We run the online self-play learner (Algorithm 1) on 2-rank Rhode Island Hold'em with $\varepsilon = 50$ and random community cards.

Table 4 shows rapid convergence: the learner discovers all 3 natural clusters by game 100 (from 100 games with random community cards), and cache hit rate reaches 99.5% by game 600. With fixed community cards, convergence is even faster (same 3 clusters by game 50), confirming that the smaller strategic space is easier to learn.

6.6 Distance Memoization

The recursive structure of RBM creates many redundant subtree comparisons. Memoizing subtree distances yields significant speedups.

On an 8-tree benchmark (2-rank deck, 28 pairwise comparisons), memoized distance computation achieves a **94× speedup** over the naive implementation, with a memo cache hit rate of approximately 97%. This demonstrates that game trees share substantial subtree structure that memoization exploits.

7 Related Work

Tree edit distance. Zhang and Jiang [1994] and Zhang [1996] established that unrestricted edit distance on unordered labeled trees is MAX SNP-hard, motivating constrained variants computable in polynomial time. Kuboyama [2007] developed tractable variations using maximum weighted bipartite matching—essentially the same core operation as RBM. These works establish that matching-based tree distances are tractable and metrizable but do not consider game-theoretic applications.

Optimal transport on trees. Mémoli [2011] defines distances between metric measure spaces using optimal couplings; RBM can be viewed as a discrete, tree-structured specialization of the Gromov-Wasserstein framework. The Tree Mover's Distance [Chuang et al., 2022] extends EMD to multisets of trees for GNN analysis. Yamada et al. [2018] formulate EMD on rooted labeled unordered trees built from complete subtrees. These works share the recursive optimal transport flavor but target different applications.

Game abstraction in poker. Gilpin and Sandholm [2007] introduced automated abstraction using k-means clustering with EMD on hand-strength histograms—flat abstraction that clusters on scalar distributions at each decision point. Ganzfried and Sandholm [2014] extended this with potential-aware EMD accounting for future card draws. Kroer and Sandholm [2014, 2018] provide the most theoretically rigorous bounds on exploitability introduced by abstraction, proving that level-by-level distance metrics control solution quality. However, their distance operates level-by-level, not recursively through the full subtree. RBM is strictly more expressive but comes with the location problem as additional overhead.

Online learning in games. CFR [Zinkevich et al., 2007] and its variants (CFR+, Monte Carlo CFR) achieve regret bounds for extensive-form games. AlphaZero [Silver et al., 2018] uses self-play with neural function approximation but lacks formal error bounds on the learned abstraction. MCTS [Coulom, 2006, Kocsis and Szepesvári, 2006] samples continuations to estimate node values—our online learner samples continuations to estimate node *identity* (location in the compressed graph). Lipschitz bandits [Kleinberg, 2005, Bubeck et al., 2011] exploit metric structure for faster learning; our setting derives the metric from game structure rather than assuming it.

The gap filled by this work. The existing literature has (1) matching-based tree distances with metric proofs, and (2) game tree abstraction with exploitability bounds. What was missing is the bridge: using a full recursive tree distance as the abstraction metric, proving that the distance bounds EV error, identifying the location problem as the fundamental complexity barrier, and the online learning formulation that dissolves it.

8 Conclusion

We have introduced the recursive bipartite matching distance, a metric on game trees that captures structural similarity—not just expected values, but the shape of the decision landscape. The metric induces a natural compression (the EV graph) with provable EV error bounds, and an online learner that builds this compression incrementally through self-play with bounded regret.

The 7–0 victory of RBM over EMD on Rhode Island Hold'em validates the central thesis: structural tree distance is a better abstraction metric than scalar hand-strength distributions.

The zero-exploitability results at natural cluster boundaries confirm that RBM identifies genuinely equivalent strategic states.

Several directions remain open. *Scalability:* the $O(n^2)$ distance computation limits application to large games; approximate matching via Sinkhorn distances [Cuturi, 2013] or learned embeddings could help while potentially preserving error bounds. *Per-level palettes:* different game stages may benefit from different clustering thresholds. *Beyond poker:* the framework applies to any extensive-form game with tree-structured state spaces, including Stratego, Magic: The Gathering, and negotiation games. *Opponent modeling:* building an RBM-based EV graph of the opponent's play patterns could enable principled exploitation.

Acknowledgments. To be added.

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A Full Proof: RBM Distance Is a Metric (Theorem 2.2)

Theorem (Restated)**.** *Let $\mathcal{T}$ be the space of rooted trees with real-valued leaves. Define $d : \mathcal{T} \times \mathcal{T} \to \mathbb{R}_{\geq 0}$ as in Definition 2.1, with leaf distance d_L and phantom penalty δ. Suppose:*

1. *d_L is a metric on the leaf value space.*
2. *$\delta(S) \geq 0$ for all subtrees S.*
3. ***Phantom triangle property:** for any subtrees S_1, S_2, $d(S_1, S_2) \leq \delta(S_1) + \delta(S_2)$.*

Then d is a metric on $\mathcal{T}$.

Proof. We verify each axiom by structural induction on the maximum depth k of the trees involved. The inductive hypothesis is that d restricted to trees of depth $< k$ satisfies all three metric axioms. The base case ($k = 0$, both trees are leaves) is immediate because d reduces to d_L, which is a metric by assumption.

Axiom 1: Identity of indiscernibles — $d(T, T) = 0$. Let T have children $\{c_1, \ldots, c_m\}$. The cost matrix W for matching T against itself has entries $W[i, j] = d(\text{sub}(c_i), \text{sub}(c_j))$. The child sets are identical and have the same cardinality, so no phantom padding is needed. The identity matching $M^* = \{(c_i, c_i)\}_{i=1}^m$ has cost:

$$\text{cost}(M^*) = \sum_{i=1}^{m} d(\text{sub}(c_i), \text{sub}(c_i)) = 0,$$

where each term vanishes by the inductive hypothesis. Since the Hungarian method finds the minimum-cost matching and $\text{cost}(M^*) = 0$ with all costs nonneg, $d(T, T) = 0$.

Conversely, if $d(T_1, T_2) = 0$, then every matched pair (c_1^i, c_2^j) has $d(\text{sub}(c_1^i), \text{sub}(c_2^j)) = 0$ (nonneg terms summing to zero), and there are no phantom-matched children (since $\delta(S) \geq 0$ and any phantom match would force the sum above zero, unless $\delta(S) = 0$, corresponding to an empty subtree). By induction, each matched pair of subtrees is identical, so $T_1 = T_2$. □

Axiom 2: Symmetry — $d(T_1, T_2) = d(T_2, T_1)$. The cost matrix W' for $d(T_2, T_1)$ is the transpose of W for $d(T_1, T_2)$:

- Real-real entries: $d(\text{sub}(c_2^j), \text{sub}(c_1^i)) = d(\text{sub}(c_1^i), \text{sub}(c_2^j))$ by inductive symmetry.
- Phantom entries: δ depends only on the real subtree, not which side it appears on.

The minimum-cost perfect matching is invariant under transposition. □

Axiom 3: Triangle inequality — $d(T_1, T_3) \leq d(T_1, T_2) + d(T_2, T_3)$. Fix three trees T_1, T_2, T_3 with children $\{a_1, \ldots, a_p\}$, $\{b_1, \ldots, b_q\}$, $\{c_1, \ldots, c_r\}$. Let M_{12} achieve cost $d(T_1, T_2)$ and M_{23} achieve cost $d(T_2, T_3)$.

Step 1: Construct a feasible matching M_{13}. Compose through T_2. Each b_j appears once in M_{12} and once in M_{23}. This creates:

- *Through-matched:* $a_i \xrightarrow{M_{12}} b_j \xrightarrow{M_{23}} c_k$. Assign $(a_i, c_k) \in M_{13}$.
- *Left-dangling:* $a_i \to b_j$ but $b_j \to \phi$. Assign $a_i \to \phi$ in M_{13}.
- *Right-dangling:* $\phi \to b_j \to c_k$. Assign $c_k \to \phi$ in M_{13}.
- *Both-phantom:* $a_i \to \phi$ in M_{12} (not through any b_j). Assign $a_i \to \phi$ in M_{13}. Symmetrically for $\phi \to c_k$ in M_{23}.

Step 2: Bound cost. For through-matched pairs, by the inductive triangle inequality:

$$d(\text{sub}(a_i), \text{sub}(c_k)) \leq d(\text{sub}(a_i), \text{sub}(b_j)) + d(\text{sub}(b_j), \text{sub}(c_k)).$$

For left-dangling a_i (via b_j matched to phantom in M_{23}): the cost in M_{13} is $\delta(\mathrm{sub}(a_i))$. The corresponding costs in M_{12} and M_{23} are $d(\mathrm{sub}(a_i), \mathrm{sub}(b_j))$ and $\delta(\mathrm{sub}(b_j))$. The 1-Lipschitz condition $\delta(\mathrm{sub}(a_i)) \leq d(\mathrm{sub}(a_i), \mathrm{sub}(b_j)) + \delta(\mathrm{sub}(b_j))$ ensures the bound holds. Right-dangling and both-phantom cases are symmetric.

Step 3: Sum. Every edge of M_{12} and M_{23} appears exactly once on the right-hand side:

$$\mathrm{cost}(M_{13}) \leq \mathrm{cost}(M_{12}) + \mathrm{cost}(M_{23}) = d(T_1, T_2) + d(T_2, T_3).$$

Since $d(T_1, T_3) \leq \mathrm{cost}(M_{13})$ by optimality of the Hungarian matching:

$$d(T_1, T_3) \leq d(T_1, T_2) + d(T_2, T_3). \quad \square$$

$\square$

Remark A.1. *With $\delta(S) = d(S, \emptyset)$, the phantom triangle property and 1-Lipschitz condition are both instances of the triangle inequality: $d(S_1, S_2) \leq d(S_1, \emptyset) + d(\emptyset, S_2) = \delta(S_1) + \delta(S_2)$ and $d(S_1, \emptyset) \leq d(S_1, S_2) + d(S_2, \emptyset)$. The theorem is self-contained.*

B Full Proof: EV Error Bound (Theorem 3.2)

Theorem (Restated)**.** *Let T_1, T_2 be rooted trees with real-valued leaves, $d(T_1, T_2) = \varepsilon$. Let $T^* = \mathrm{merge}(T_1, T_2)$ with equal-weight leaf averaging. Then $|\mathrm{EV}(T^*) - \mathrm{EV}(T_i)| \leq \varepsilon/2$ for $i \in \{1, 2\}$.*

Proof. By structural induction on tree depth.

Base case: leaves. $T_1 = \mathrm{leaf}(v_1)$, $T_2 = \mathrm{leaf}(v_2)$, $d(T_1, T_2) = |v_1 - v_2| = \varepsilon$. The merge is $T^* = \mathrm{leaf}((v_1 + v_2)/2)$, so:

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_1)| = \left| \frac{v_1 + v_2}{2} - v_1 \right| = \frac{|v_2 - v_1|}{2} = \frac{\varepsilon}{2}. \quad \checkmark$$

Inductive case: internal nodes. Let T_1 have children $\{a_1, \ldots, a_m\}$ and T_2 have children $\{b_1, \ldots, b_n\}$ with $m \leq n$. Let M^* be the optimal matching with matched-pair costs $d_i = d(\mathrm{sub}(a_i), \mathrm{sub}(b_{\sigma(i)}))$ and phantom costs $\delta(b_j)$ for $j \notin \mathrm{im}(\sigma)$:

$$\varepsilon = \sum_{i=1}^{m} d_i + \sum_{j \notin \mathrm{im}(\sigma)} \delta(b_j).$$

Case 1: Phantom children dropped (lossy merge). With uniform weighting:

$$\begin{aligned}
|\mathrm{EV}(T^*) - \mathrm{EV}(T_1)| &= \left| \frac{1}{m} \sum_{i=1}^{m} \mathrm{EV}(c_i^*) - \frac{1}{m} \sum_{i=1}^{m} \mathrm{EV}(\mathrm{sub}(a_i)) \right| \\
&\leq \frac{1}{m} \sum_{i=1}^{m} |\mathrm{EV}(c_i^*) - \mathrm{EV}(\mathrm{sub}(a_i))| \\
&\leq \frac{1}{m} \sum_{i=1}^{m} \frac{d_i}{2} \\
&= \frac{1}{2m} \sum_{i=1}^{m} d_i \leq \frac{\varepsilon}{2},
\end{aligned}$$

where the third line uses the inductive hypothesis on each matched pair, and the last step uses $\sum_i d_i \leq \varepsilon$.

Equal-structure case ($m = n$, no phantoms). The bound tightens:

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_1)| \leq \frac{1}{2m} \sum_{i=1}^{m} d_i = \frac{\varepsilon}{2m}.$$

For $m \geq 2$, this is at most $\varepsilon/4$, strictly better than the stated bound.

The bound $\varepsilon/2$ is tight only at leaves ($m = 0$) or degenerate single-child chains ($m = 1$ at every internal node). □

Corollary (Weighted Merges)**.** *For weights w_1, w_2 with $w_1 + w_2 = 1$, merged leaf values are $v^* = w_1 v_1 + w_2 v_2$. The error bound generalizes to:*

$$|\mathrm{EV}(T^*) - \mathrm{EV}(T_i)| \leq w_{3-i} \cdot \varepsilon.$$

For $w_1 = w_2 = 1/2$, this recovers $\varepsilon/2$. For $w_1 \to 1$, the error for T_1 vanishes while that for T_2 approaches ε.

C Full Proof: Online Regret Bound (Theorem 5.1)

Theorem (Restated)**.** *Consider the online EV graph learner with threshold $\varepsilon > 0$, playing T games against a stationary environment. Let K be the number of strategic equivalence classes and $V_{\max} = \max_t |\mathrm{EV}(T_t)|$. Then:*

1. *Exploration bound: total cache misses $\leq K$.*
2. *Per-step exploitation error: at most $\varepsilon/2$.*
3. *Cumulative regret: $R(T) \leq K V_{\max} + (T - K)\varepsilon/2$.*
4. *Average regret: $R(T)/T \to \varepsilon/2$ as $T \to \infty$.*
5. *With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$: $R(T) = O(\sqrt{T})$.*

Proof. **Part 1: Exploration bound.** Define a strategic equivalence class as a maximal set $\{T : d(T, T') \leq \varepsilon\}$ centered at some T'. Since the game is finite, the number of such classes under fixed $\varepsilon > 0$ is some finite K.

A cache miss occurs when T_t lies outside all existing clusters: $d(T_t, \mathrm{rep}(c)) > \varepsilon$ for every cluster c. A new cluster is created with T_t as representative.

Each equivalence class generates at most one miss: once $T_t \in \mathcal{C}$ triggers a miss, the new cluster covers all future $T_{t'} \in \mathcal{C}$ since $d(T_{t'}, T_t) \leq \varepsilon$ by definition. Therefore total misses $\leq K$. □

Part 2: Per-step exploitation error. When exploiting cluster c^* on game tree T_t with $d(T_t, \mathrm{rep}(c^*)) \leq \varepsilon$, the cluster representative serves the role of the merged tree. By Theorem 3.2:

$$r_t = |\mathrm{EV}(\mathrm{rep}(c^*)) - \mathrm{EV}(T_t)| \leq \frac{d(T_t, \mathrm{rep}(c^*))}{2} \leq \frac{\varepsilon}{2}.$$

Part 3: Cumulative regret bound. Partition the T games into exploration games $\mathcal{E}$ (cache misses, $|\mathcal{E}| \leq K$) and exploitation games $\mathcal{X}$ (cache hits, $|\mathcal{X}| \geq T - K$).

Exploration: worst-case per-game regret is $V_{\max}$. Summing: $\sum_{t \in \mathcal{E}} r_t \leq K V_{\max}$.

Exploitation: per-game regret $\leq \varepsilon/2$. Summing: $\sum_{t \in \mathcal{X}} r_t \leq (T - K)\varepsilon/2$.

Total:

$$R(T) = \sum_{t \in \mathcal{E}} r_t + \sum_{t \in \mathcal{X}} r_t \leq K V_{\max} + (T - K)\frac{\varepsilon}{2}.$$

Part 4: Average regret convergence.

$$\frac{R(T)}{T} \leq \frac{KV_{\max}}{T} + \frac{(T-K)}{T} \cdot \frac{\varepsilon}{2} \xrightarrow{T\to\infty} \frac{\varepsilon}{2}.$$

The exploration cost $KV_{\max}$ is a one-time overhead amortized over all future games. After all K classes are discovered, every subsequent game is pure exploitation with per-game regret at most $\varepsilon/2$. □

Part 5: Sublinear regret via ε-annealing. Let $\varepsilon(t) = \varepsilon_0/\sqrt{t}$. Let $K(\varepsilon)$ denote the number of equivalence classes at threshold ε, bounded by N (total distinct game trees).

Total exploration events across all T rounds is at most $K(\varepsilon(T)) \leq N$, since each decrease in ε that splits a class triggers at most one new exploration event.

Exploitation regret at time t is at most $\varepsilon(t)/2 = \varepsilon_0/(2\sqrt{t})$. Summing:

$$\sum_{t\in\mathcal{X}} \frac{\varepsilon_0}{2\sqrt{t}} \leq \sum_{t=1}^{T} \frac{\varepsilon_0}{2\sqrt{t}} \leq \frac{\varepsilon_0}{2} \cdot 2\sqrt{T} = \varepsilon_0\sqrt{T},$$

using the standard bound $\sum_{t=1}^{T} t^{-1/2} \leq 2\sqrt{T}$.

Adding exploration cost:

$$R(T) \leq NV_{\max} + \varepsilon_0\sqrt{T} = O(\sqrt{T}).$$

Average regret:

$$\frac{R(T)}{T} \leq \frac{NV_{\max}}{T} + \frac{\varepsilon_0}{\sqrt{T}} \to 0 \quad \text{as } T \to \infty.$$ □

Remark C.1 (Comparison with Standard Bandit Bounds)**.** *The $O(\sqrt{T})$ rate under ε-annealing matches the minimax-optimal rate for multi-armed bandits with K arms [Auer et al., 2002]. The EV graph learner is effectively a bandit over clusters, with the RBM distance providing a geometry on the arm space: nearby arms have similar payoffs by the EV error bound. This is analogous to Lipschitz bandits [Kleinberg, 2005], where metric structure enables faster learning. In the EV graph setting, the metric is not assumed but derived from the game structure.*

Remark C.2 (Tightness)**.** *The exploration term $KV_{\max}$ is tight when each class is first encountered at a worst-case game with adversarial exploration (zero payoff). In practice, heuristic strategies achieve reasonable EV, so the effective exploration cost is much lower. The exploitation term $(T-K)\varepsilon/2$ is tight at leaves; for trees with branching factor $m > 1$, Theorem 3.2 gives a tighter per-step bound of $\varepsilon/(2m)$.*